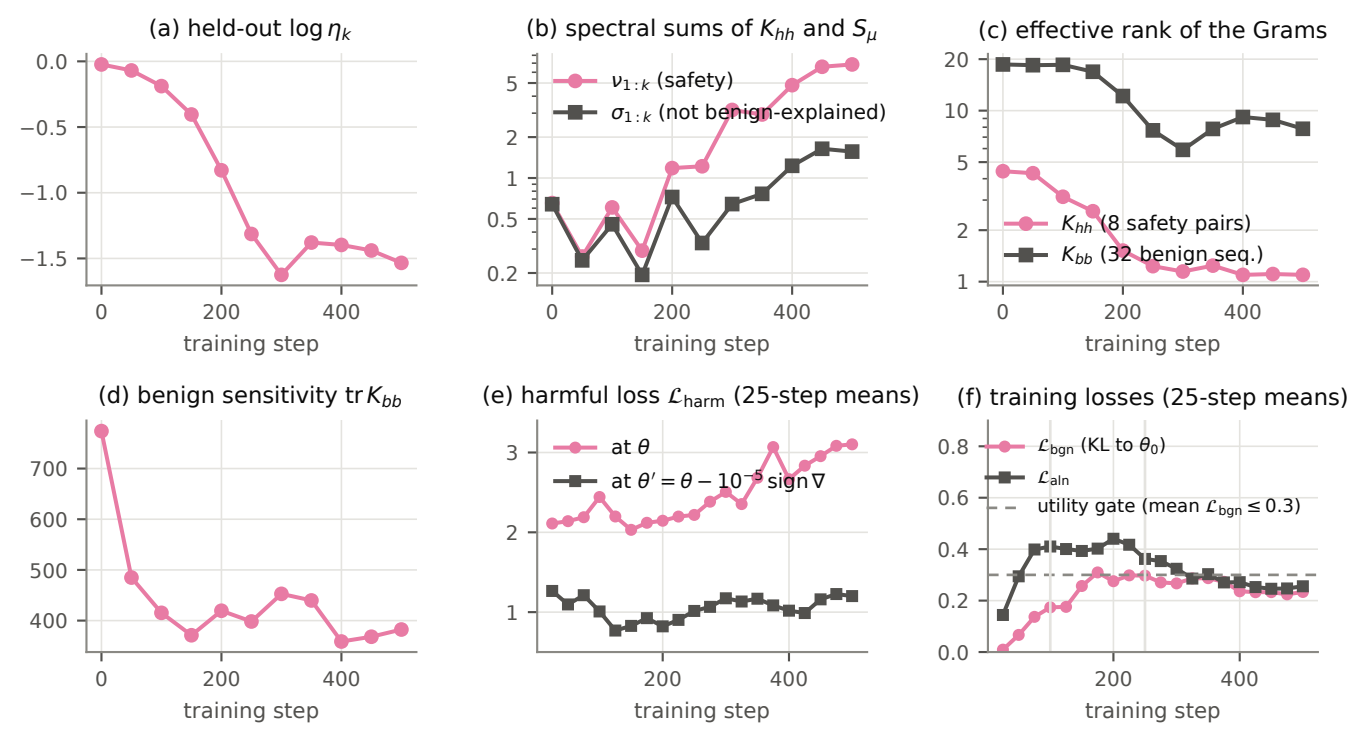

(a) held-out log η_k
0.0
−0.5
−1.0
−1.5
0
200
400
training step
(b) spectral sums of K_{hh} and S_μ
$\nu_{1:k}$ (safety)
$\sigma_{1:k}$ (not benign-explained)
5
2
1
0.5
0.2
0
200
400
training step
(c) effective rank of the Grams
20
10
5
2
1
K_{hh} (8 safety pairs)
K_{bb} (32 benign seq.)
0
200
400
training step
(d) benign sensitivity tr K_{bb}
700
600
500
400
0
200
400
training step
(e) harmful loss $\mathcal{L}_{\text{harm}}$ (25-step means)
at θ
at $\theta' = \theta - 10^{-5}\,\text{sign}\,\nabla$
3
2
1
200
400
training step
(f) training losses (25-step means)
$\mathcal{L}_{\text{bgn}}$ (KL to θ_0)
$\mathcal{L}_{\text{aln}}$
utility gate (mean $\mathcal{L}_{\text{bgn}} \le 0.3$)
0.8
0.6
0.4
0.2
0.0
200
400
training step